

FUKUOKA AP, Japan

WMO: 478080

Lat: 33.584N

Lon: 130.452E

Elev: 12

StdP: 101.18

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	1.0	-8.9	1.8	4.0	-7.2	2.0	4.6	10.1	4.4	9.3	5.3	3.4	150	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	34.0	25.7	33.0	25.5	32.0	25.1	26.7	31.4	26.3	30.8	25.9	30.2	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.4	29.1	25.0	20.1	28.9	24.2	19.2	28.6	83.9	31.8	81.8	31.2	79.9	30.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.2	7.4	-2.2	35.8	1.0	0.8	-3.0	36.4	-3.6	36.9	-4.2	37.4	-4.9	38.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-3.8	27.4	1.2	0.7	-4.7	27.9	-5.4	28.3	-6.0	28.7	-6.9	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.2	7.2	10.4	15.2	19.9	23.5	27.7	28.7	24.8	19.5	13.8	8.5
	DBStd	8.16	2.57	3.07	3.14	3.06	2.41	2.24	2.31	1.90	2.55	2.71	3.17	2.91
	HDD10.0	313	121	88	32	2	0	0	0	0	0	0	4	66
	HDD18.3	1509	376	311	246	101	11	0	0	0	0	18	140	306
	CDD10.0	2935	3	11	45	158	308	407	550	579	444	294	118	18
	CDD18.3	1089	0	0	0	7	61	157	292	321	194	54	3	0
	CDH23.3	9903	0	0	1	26	274	911	3174	3791	1511	211	6	0
	CDH26.7	3563	0	0	0	2	34	202	1261	1631	413	20	0	0
Wind	WSAvg	3.5	3.7	3.6	3.8	3.7	3.5	3.2	3.6	3.6	3.1	3.0	3.2	3.6
Precipitation	PrecAvg	1647	73	70	100	125	142	254	266	184	180	95	87	64
	PrecMax	2980	208	155	224	240	496	642	886	847	609	290	182	148
	PrecMin	895	13	18	21	35	37	15	19	8	24	11	13	16
	PrecStd	369	38	33	42	52	80	133	182	144	110	63	44	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.2	22.1	26.0	29.2	32.1	34.9	35.9	33.2	28.8	24.1	18.2
		MCWB	12.5	14.5	13.8	18.0	18.8	24.4	26.0	25.9	25.2	22.2	17.7	13.5
	2%	DB	13.9	16.2	19.8	23.9	27.8	30.1	33.8	34.2	31.9	27.0	22.0	16.2
		MCWB	10.1	12.3	13.4	16.6	18.7	23.1	25.7	25.7	24.7	20.6	16.7	11.9
	5%	DB	12.1	14.1	17.8	22.1	26.1	29.0	32.9	33.2	30.2	25.8	20.2	15.0
		MCWB	8.4	10.2	12.7	15.3	17.9	22.4	25.5	25.6	23.8	19.8	15.5	11.0
	10%	DB	11.0	12.8	16.0	20.8	24.9	27.8	31.9	32.8	29.0	24.2	19.1	13.9
		MCWB	7.4	9.0	11.5	14.8	17.8	22.0	25.2	25.4	23.2	18.6	14.9	10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	15.5	16.6	20.6	22.7	25.8	27.2	27.6	26.4	23.6	19.4	15.2
		MCDB	15.5	17.9	19.0	23.2	25.4	29.3	32.1	32.4	30.4	26.6	22.0	16.7
	2%	WB	11.3	13.2	14.9	18.2	21.5	24.9	26.7	26.9	25.6	22.3	17.8	13.1
		MCDB	13.1	15.9	18.0	21.7	24.3	28.3	31.4	31.7	29.5	25.7	20.4	15.2
	5%	WB	9.2	11.2	13.3	16.9	20.2	24.1	26.2	26.5	24.9	21.0	16.7	11.7
		MCDB	11.4	13.3	16.6	20.3	23.6	27.5	30.7	31.2	28.8	24.1	19.6	14.2
	10%	WB	7.7	9.5	12.2	15.7	19.1	23.2	25.8	26.0	24.2	19.7	15.5	10.4
		MCDB	10.3	12.1	15.6	19.1	22.9	26.4	30.1	30.7	28.0	23.0	18.6	13.2
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.2	7.8	8.3	8.2	6.6	6.3	6.8	6.9	7.8	7.8	7.1
		MCDBR	9.0	9.8	10.5	10.6	10.4	8.4	7.5	8.0	8.0	9.1	9.0	8.9
		MCWBR	6.4	6.9	6.4	5.5	4.6	3.6	2.8	2.9	3.3	4.3	5.1	6.0
	5% WB	MCDBR	7.7	8.0	8.4	7.6	7.2	6.1	6.8	7.0	6.9	6.5	7.3	7.5
		MCWBR	6.6	6.9	6.3	5.3	4.8	3.5	2.9	3.0	3.2	3.9	5.1	6.1
Clear-Sky Solar Irradiance	taub	0.433	0.474	0.536	0.556	0.563	0.578	0.570	0.537	0.507	0.461	0.433	0.425	
	taud	2.047	1.973	1.827	1.833	1.875	1.923	1.989	2.077	2.112	2.141	2.132	2.082	
	Ebn at Noon	744	756	742	752	751	738	741	759	762	763	742	725	
	Edh at Noon	135	161	201	208	202	192	179	161	149	134	123	123	
All-Sky Solar Radiation	RadAvg	2.17	2.88	3.97	4.80	5.21	4.44	4.84	5.10	4.10	3.48	2.58	2.01	
	RadStd	0.21	0.28	0.27	0.41	0.60	0.55	0.72	0.56	0.42	0.36	0.24	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page